

Paper B and Abramsky–Cerecescu–Constantin: an explicit equivalence, and what is new

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July 2026

Abstract

The value classification of state-independent qudit contextuality — the even/odd dichotomy and the fixed obstruction value $d/2$ — is due to Abramsky, Cerecescu and Constantin [1]. Paper B of this program re-derives it in the Weyl–Heisenberg realization. This note makes that relationship *precise*: we give an object dictionary, prove that the Weyl commutator carry is their commutation factor under it, and conclude that Paper B’s Obstruction Spectrum Theorem is the Weyl specialization of their Theorem 16 — neither independent of nor a strengthening of it. We then delimit exactly the content that is *not* in [1]. Every correspondence claimed here is machine-checked in `verification/acc_correspondence.py` (PM, and certificates at $d = 2, 8, 16$).

1 The two frameworks

Abramsky–Cerecescu–Constantin [1]. A *commutation group* $G(\mu)$ is presented by generators $x_1, \dots, x_n$ over $\mathbb{Z}_d$ with a commutator matrix $\mu(x, y) = v_{x,y}$ and a central phase group $\mathbb{Z}_d$. A *contextual word* (w, β, k) is a bracketed word of commuting subproducts whose evaluated central phase is $k \neq 0$; it witnesses state-independent contextuality (their Thm. 9: one exists iff $G(\mu)$ is state-independently contextual). Their *inversion sum* $\sum \mathcal{I}(w)$ records the central phase accrued in reducing w to normal form, and (their Lemma 15) for a word of commuting products

$$2 \sum \mathcal{I}(w) = \sum_{x < y} n_x n_y v_{y,x}, \quad (1)$$

n_x the multiplicity of generator x . Their Theorem 16 concludes: for a contextual word, $2k \equiv 0 \pmod{d}$ with $k = \sum \mathcal{I}(w)$, so d must be even and the only nonzero value is $d/2$; their Theorem 18 gives noncontextuality for odd d ; Theorem 21 classifies the Darboux-normal-form matrices that admit contextual words.

Paper B (Weyl realization). Observables are symmetric Weyl operators $W(v) = \tau^{-q(v)} X^{a_1} Z^{b_1} \otimes \dots$, $\tau = e^{i\pi/d}$, with symplectic form $\langle u, v \rangle = \sum_i (u_{a,i} v_{b,i} - u_{b,i} v_{a,i})$. A *context* C is a commuting, sum-zero word; its product is $\omega^{s(C)} I$, $\omega = e^{2\pi i/d}$. A *certificate* λ ($\lambda^\top A \equiv 0$) has value $S = \lambda^\top s \pmod{d}$, and the *commutator carry* is $b(C) = \sum_{i < j} \langle c_i, c_j \rangle_{\mathbb{Z}/d} \pmod{2}$. Theorem 1 (Obstruction Spectrum): $2S \equiv 0 \pmod{d}$, so $S \in \{0, d/2\}$ (even d), $\{0\}$ (odd d).

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`verification/acc_correspondence.py`.

2 The dictionary

<i>Paper B (Weyl)</i>	<i>ACC [1]</i>
Weyl label v	commutation-group generator/vector
symplectic form $\langle u, v \rangle$	commutator matrix entry $\mu(x, y) = v_{x,y}$
closed context C	commuting-product (sub)word
certificate λ	contextual word (w, β, k)
obstruction value S	central phase k
commutator carry $b(C)$	parity of the commutation factor $\sum \mathcal{I}(w)$
$\mathbb{Z}_d$ value assignment	left splitting / noncontextual homomorphism

Proposition 1 (Carry = commutation factor). *Under the dictionary, the Weyl integer pair-sum of a context equals the right-hand side of the ACC identity (1):*

$$\sum_{i < j} \langle c_i, c_j \rangle_{\mathbb{Z}} = \sum_{x < y} n_x n_y \langle x, y \rangle_{\mathbb{Z}} = 2 \sum \mathcal{I}(w).$$

Hence $b(C)$ is the mod-2 reduction of $\sum \mathcal{I}(w)$, and the Weyl polarization step in the proof of Theorem 1 is exactly ACC’s Lemma 15.

Proof. The first equality is the standard expansion of a sum over an ordered multiset into distinct-generator multiplicity products (self-pairs contribute $\langle x, x \rangle = 0$); the second is (1) with $v_{y,x} = \langle x, y \rangle_{\mathbb{Z}}$. Machine-verified on the six Peres–Mermin contexts and on random closed contexts at $d = 2, 4, 6, 8$ in `acc_correspondence.py` (all “[II identical] True”). $\square$

Theorem 2 (Paper B Thm. 1 is the Weyl specialization of ACC Thm. 16). *For any Weyl context family, the certificate value S equals the ACC central phase k of the corresponding contextual word, and Theorem 1’s conclusion $2S \equiv 0 \Rightarrow S \in \{0, d/2\}$ is Theorem 16’s conclusion $2k \equiv 0 \Rightarrow k \in \{0, d/2\}$. In particular Paper B’s Theorem 1 is neither independent of, nor a strengthening of, ACC’s Theorem 16: it is that theorem realized in the Weyl–Heisenberg group.*

Proof. By Proposition 1 the two derivations share the polarization identity, and the Weyl telescoping $2s(C) \equiv -2 \sum_{u \in C} q(u)$ paired against a certificate ($\lambda^T A \equiv 0$) gives $2S \equiv 0$, matching $2k \equiv 0$; the carried value is $S = k$. The certificate values realized on concrete AvN witnesses are $S = k = d/2$: Peres–Mermin ($S = 1$), the ROZF $d = 8$ certificate ($S = 4$), and the fibered $d = 16$ certificate ($S = 8$), machine-checked in `acc_correspondence.py`. $\square$

3 What is new (and what is not)

Not new. The even/odd dichotomy, the value $\{0, d/2\}$, the parity mechanism $2k \equiv 0$, and the classification of which families are contextual are ACC’s (Thms. 16, 18, 21). The carry $b(C)$ is their commutation factor in symplectic-form coordinates: a change of realization and an explicit closed form, not a new invariant. Their $\mathbb{Z}_2 \rightarrow \mathbb{Z}_{2k}$ transfer construction is prior art for lifting witnesses up the tower.

Candidate new content (offered at exactly this strength):

1. an *explicit value-bit closed form* for 2-adically lifted certificates (Paper B, Thm. W), a computation ACC do not perform (they have no 2-adic lift);

2. the *cross-dimensional* tower statements — Depth Rigidity and the fiber doubling law — which are theorems about relations *between* d and $2d$, and are new only insofar as they are not corollaries of “the only nonzero value is $d/2$ ” restated at each d ; the ACC transfer is their baseline;
3. a three-device superconducting replication of the $d = 2$ witness (statistical, assumption-dependent — see the repository’s hardware notes);
4. the *geometric/cohomological home*: the identification of the obstruction with a canonical degree-2 class of order n on the stack $[X/\mathrm{PU}(n)]$ of maximal abelian subalgebras (companion paper), which ACC’s combinatorial (rewriting-system) treatment does not address. This is the program’s distinctive contribution.

Honest status. Items 1 and 3 are established at the stated (computational / experimental) strength. Item 2 requires each tower statement to be proved genuinely cross-dimensional; that separation is the natural next step. Item 4 is the subject of the companion paper and awaits specialist certification of its Morita / spectral-sequence bookkeeping.

References

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- [3] R. Raussendorf, C. Okay, M. Zurel, P. Feldmann, *The role of cohomology in quantum computation with magic states*, Quantum **7**, 979 (2023).